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Complex-valued neural networks for spectral induced polarization applications

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SUMMARY

Spectral induced polarization (SIP) aims to characterize geological materials by measuring the dispersion of their complex conductivity in the frequency domain. Despite the complex-valued nature of SIP data, most machine learning models used for its analysis rely on real-valued representations that discard phase information and may limit performance. This study investigates the benefits of complex-valued neural networks (CVNN) for SIP applications by comparing their performance against real-valued neural networks (RVNN) across three tasks: mineral classification, Cole–Cole parameter estimation, and mechanistic modelling of ionic and electric potential perturbations around polarizable minerals. To ensure fair comparisons and emphasize the effect of complex-valued representations, we design CVNN and RVNN models with matched capacity, aspect ratio and training duration. Our numerical experiments show that CVNNs consistently outperform RVNNs in the classification task, achieving lower validation loss and up to 5 per cent higher classification metrics (p-value $= 2.9 \times 10^{-7}$). We test the Cole–Cole inversion networks on laboratory SIP measurements and validate the parameter estimation accuracy using synthetic data. Test results indicate that CVNNs produce curve fits that are ≈ 4 % more accurate for the imaginary part of resistivity (p-value $= 3.1 \times 10^{-4}$), and validation results show accuracy improvements of up to 2 per cent for chargeability, relaxation time and the Cole–Cole exponent (p-value $= 1.7 \times 10^{-7}$). CVNNs also yield more accurate approximations of mechanistic model variables, with error reductions of up to 1 per cent for ionic concentrations (p-value $= 1.6 \times 10^{-4}$). Our experiments suggest that CVNNs provide modest but statistically significant benefits in SIP applications involving laboratory or synthetic data. While RVNNs may eventually reach comparable predictive accuracy if trained longer, we observe that CVNNs converge more rapidly under matched training conditions. This study provides a reproducible framework for benchmarking neural network architectures in SIP and supports the integration of CVNNs into geophysical workflows where phase responses encode physically meaningful information.

Key words: Electrical properties; Machine learning; Neural networks, fuzzy logic; Numerical modelling.

1 INTRODUCTION

Spectral induced polarization (SIP) is a widely used geophysical method for characterizing the electrical properties of rocks and soils. SIP quantifies both the resistive and capacitive response of geological materials subjected to a variable electric field by measuring their complex conductivity, which depends on pore structure (Kruschwitz *et al.* 2010), interstitial fluid properties (Bairlein *et al.* 2016) and mineral composition and texture (Bérubé *et al.* 2019; Gurin *et al.* 2021). Unlike the direct current resistivity method, SIP explicitly captures frequency-dependent polarization effects associated with electrochemical interactions at mineral-fluid interfaces, which occur at various characteristic timescales. This capability makes SIP valuable for distinguishing rock types (Lockner & Byerlee 1985), assessing mineralization (Nelson & Voorhis 1983; Revil *et al.* 2022), monitoring subsurface contamination (Mewafy *et al.* 2013), and hydrogeophysical applications (Roudsari *et al.* 2024). As a result, researchers and practitioners commonly use SIP in

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exploration geophysics, environmental studies and hydrogeology, where characterizing the electrical properties of rocks and soils can inform decisions regarding resource management and environmental monitoring.

Quantitatively interpreting the SIP signatures of geomaterials is challenging because existing modelling approaches often oversimplify the underlying physics or fail to capture the intricacies of mineralogical, textural and fluid-related responses. Mechanistic complex conductivity models (e.g. Revil *et al.* 2015; Misra *et al.* 2016; Bücker *et al.* 2018; Jin *et al.* 2019) suffer from overparametrization and imbalanced parameter sensitivity, making them difficult to calibrate and unreliable for estimating geomaterial properties based on SIP data (Bérubé & Baron 2023). Alternatively, equivalent circuit models provide robust fits to SIP data but rely on parameters that lack proper physical meaning in the context of geomaterial characterization (e.g. Dias 2000; Shin *et al.* 2015). Furthermore, establishing correlations between geological properties and the parameters of empirical models (e.g. Cole–Cole or Debye decomposition) is often ambiguous. This ambiguity arises because multiple factors, such as pore space characteristics (Koch *et al.* 2011, 2012), pore water salinity (Wu & Peruzzo 2020), saturation (Martin *et al.* 2021) and grain size (Gurin *et al.* 2018), can influence the same model parameters simultaneously, making it difficult to isolate the contribution of any single property.

We hypothesize that thanks to their ability to learn complex patterns directly from data, neural networks can offer an alternative to conventional modelling approaches to improve SIP data analysis. Real-valued neural networks (RVNNs) are widely applied in geophysics, particularly for lithology classification, forward modelling, parameter estimation and inversion tasks because they efficiently process large data sets and have inherent regularization and parallelization (e.g. Bressan *et al.* 2020; Obornev *et al.* 2020; Zhang & Curtis 2021; Bérubé & Gagnon 2025). However, when applied to complex-valued geophysical data, RVNNs may fail to capture phase information and frequency-dependent behaviour fully (Hirose 2012; Lee *et al.* 2022). As they do not inherently handle complex numbers, these networks typically rely on data transformation techniques, such as representing the real and imaginary components as independent inputs. This approach can introduce inefficiencies by doubling the input size and obscuring phase relationships, limiting the network's ability to exploit the entire data structure (Xiao *et al.* 2021). These limitations have motivated the development of complex-valued neural networks (CVNNs), which naturally process complex inputs and are well suited for applications where the coherence between amplitude and phase is essential.

SIP data consist of amplitude and phase measurements of a geomaterial's conductivity across multiple frequencies, producing inherently complex-valued numbers. In practice, laboratory SIP measurements can cover a broad frequency range (e.g. 1 mHz to 10 MHz, Volkmann & Klitzsch 2015). Field surveys, however, are often constrained by instrumental, logistical and environmental limitations, and use a more narrow frequency range (e.g. 70 mHz to 1 kHz, Flores Orozco *et al.* 2012). CVNNs extend traditional neural network architectures by incorporating complex arithmetic, weights and activation functions. Unlike RVNNs, CVNNs operate directly in the complex domain. By preserving the complex-valued structure of SIP data, CVNNs may provide a more suitable framework for representing the frequency dispersion of polarization and conduction effects characteristic of geomaterials.

CVNNs outperform RVNNs in various fields where data naturally resides in the complex domain. For example, Virtue *et al.* (2017) use a CVNN with cardioid activation functions for magnetic resonance fingerprinting, achieving greater accuracy than real-valued models in identifying tissue parameters from complex magnetic resonance imaging signals. Similarly, Dong & Huang (2020) introduce a complex-valued tree parity machine for neural cryptography and demonstrate enhanced security compared with equivalent RVNNs, and Wang *et al.* (2021) report improved accuracy and faster convergence in distinguishing wireless communication emitters with CVNNs. In image recognition, Mönning & Manandhar (2018) show that CVNNs effectively handle noise in the complex plane, though they observe that complex weight initialization remains a challenge, and Cao *et al.* (2019) use a CVNN that outperforms contemporary state-of-the-art methods in remote sensing image classification. Moreover, Barrachina *et al.* (2021a) highlight the advantages of CVNNs for non-circular complex data sets in signal processing and report reduced overfitting and narrower confidence intervals compared with real-valued models. Lastly, Gao *et al.* (2019) propose a CVNN framework for radar imaging and demonstrate improvements in imaging quality and computational efficiency.

Despite their success in other domains, applications of CVNNs in geophysics remain limited. In seismic analysis, Dramsch *et al.* (2021) show that a smaller CVNN outperforms larger real-valued models in processing non-stationary data. Similarly, Kim *et al.* (2019) and Pham *et al.* (2020) apply CVNNs to seismic noise attenuation and note improved signal preservation compared with RVNNs, and Ristea & Radoi (2022) introduce a complex-valued convolutional approach for earthquake parameter estimation and achieve high accuracy in predicting epicentral distance, depth and magnitude. Electromagnetic geophysics has also seen limited CVNN applications, with Guo *et al.* (2021) employing a complex-valued conditional generative adversarial network for electromagnetic inverse scattering, demonstrating that CVNNs outperform real-valued convolutional networks. To the best of our knowledge, no prior studies have applied CVNNs to SIP data, though Bérubé & Baron (2023) briefly discuss the possibility.

The primary objective of this study is to rigorously compare the performance of RVNNs and CVNNs in SIP applications. Our original contributions are threefold. First, we assess whether CVNNs offer a statistically significant advantage over RVNNs in SIP data analysis through controlled comparisons across three representative tasks: classifying polarizable minerals, estimating Cole–Cole parameters and approximating mechanistic SIP models. Secondly, we release a reproducible benchmark data set and evaluation framework to support future research on complex-valued machine learning of SIP. Thirdly, we demonstrate that the observed performance gains with CVNNs extend beyond synthetic benchmarks by testing our approach on SIP spectra of Canadian Malartic rock samples.

The next section provides the fundamental principles of SIP data and neural networks. We then describe the machine learning methodology, including data models and data generation strategies. A detailed presentation of the experimental setup follows, including the network configurations and performance evaluation across the three aforementioned tasks. The results section compares the strengths and limitations of each network type, and we discuss their implications for SIP applications. Finally, we summarize the study's key findings and outline potential directions for future research.

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2 THEORY

2.1 Spectral induced polarization

As measured by the SIP method, the capacitive properties of rocks and soils are assumed to be mainly associated with the surface polarization of electron-conducting minerals due to the accumulation of electrical charges at the interface between minerals and interstitial water. The formation of an electrical double layer at the mineral grain surfaces induces a voltage perturbation, which is assumed to be linear and proportional to the current normal to the grain surface (Wait 1982).

2.1.1 Complex conductivity representation

The classical measure of induced polarization is chargeability, a parameter that simplifies the frequency-dependent properties of electrical conductivity. Chargeability, m, is defined by Seigel (1959) as

$$m = \frac{\sigma_\infty - \sigma_0}{\sigma_\infty}, \tag{1}$$

where σ_0 is the direct current conductivity and σ_∞ is the instantaneous conductivity. However, SIP provides a more comprehensive representation of conductivity dispersion between σ_0 and σ_∞ by expressing data in terms of the frequency-domain effective complex conductivity, σ_{eff}, reading

$$\sigma_{\text{eff}}(\omega) = \sigma'(\omega) + i\sigma''(\omega), \tag{2}$$

where $i^2 = -1$ is the imaginary unit, $\omega = 2\pi f$ is the angular frequency and f is the excitation frequency in Hz. The complex conductivity formulation is advantageous because its real part, σ', corresponds to conduction mechanisms, whereas its imaginary part, σ'', describes temporary and reversible storage of electrical energy (Binley & Slater 2020). The complex conductivity representation also facilitates the integration of anisotropy effects in SIP modelling (Bérubé & Gagnon 2025). Alternatively, many studies express SIP data as complex resistivity, denoted by $\rho_{\text{eff}} = \sigma_{\text{eff}}^{-1}$.

2.1.2 Kramers–Kronig relations

The Kramers–Kronig (KK) relations establish a fundamental connection between the real and imaginary components of a linear causal response function. These relations allow a model to uniquely determine one component when it has access to the other across all frequencies. In the context of SIP, the relaxation phenomena associated with interfacial processes at mineral surfaces contribute to a complex-valued linear response. Consequently, the KK relations apply to the effective conductivity and resistivity describing a geomaterial's wideband response to a time-dependent excitation (Volkmann & Klitzsch 2015). According to KK relations, the Hilbert transform $\mathcal{H}$ relates the real and imaginary components of conductivity through (Bédard & Destexhe 2018)

$$\begin{aligned} \sigma'(\omega) &= \mathcal{H}(\sigma''(\omega)) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sigma''(\omega')}{\omega' - \omega}\,\mathrm{d}\omega', \\ \sigma''(\omega) &= -\mathcal{H}(\sigma'(\omega)) = -\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\sigma'(\omega')}{\omega' - \omega}\,\mathrm{d}\omega'. \end{aligned} \tag{3}$$

From a machine learning perspective, eq. (3) imposes a physics-informed constraint: among all possible models that can fit complex conductivity data, only those whose predictions satisfy the KK relations are physically valid. An alternative interpretation is that the real part of conductivity uniquely defines the imaginary part. A neural network with sufficient capacity can theoretically learn a task using σ' or σ'' as its sole input if the associated frequency range is sufficiently large. However, neural networks trained on only one component are not guaranteed to satisfy KK constraints since the learned mapping may not implicitly define the other component through the Hilbert transform but instead approximate it with an arbitrary hidden representation.

2.2 Neural networks

2.2.1 Real-valued neural networks

Consider an RVNN with L fully connected hidden layers. The RVNN approximates the output of an arbitrary function, $\mathbf{y} \in R$, given an input vector, $\mathbf{x} \in R$. The network output is given by

$$\mathbf{y} = \mathbf{W}^{(L+1)}\mathbf{z}^{(L)} + \mathbf{b}^{(L+1)}, \tag{4}$$

where the output of each lth hidden layer is

$$\mathbf{z}^{(l)} = \psi\left(\mathbf{W}^{(l)}\mathbf{z}^{(l-1)} + \mathbf{b}^{(l)}\right), \tag{5}$$

with ψ denoting a nonlinear, element-wise activation function. The weight matrices $\mathbf{W} \in R$ and bias vectors $\mathbf{b} \in R$ are trainable parameters. The input layer is $\mathbf{z}^{(0)} = \mathbf{x}$.

The universal approximation theorem states that RVNNs with at least one sufficiently large hidden layer and a nonlinear activation function can approximate any continuous function (Hornik *et al.* 1989). RVNNs leverage a range of activation functions to introduce nonlinearity, with the rectified linear unit (ReLU) being particularly popular for its numerical efficiency and lack of output saturation, contrary to sigmoid functions. Alternatively, activation functions like Swish and Mish have shown potential advantages in convergence and accuracy (Ramachandran *et al.* 2018; Misra 2020).

2.2.2 Optimization of real-valued parameters

The RVNN parameters, $\boldsymbol{\theta} = \{\mathbf{W}, \mathbf{b}\}$, are usually optimized via back-propagation using a variant of the stochastic gradient descent algorithm (e.g. Kingma & Ba 2015). The optimization process relies on computing the gradient of a loss function, $\mathcal{L}$, with respect to the RVNN parameters. The choice of loss function depends on the task: classification tasks commonly use cross-entropy loss, while regression tasks often employ mean squared error (MSE). The RVNN weights are optimized with

$$\boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}_n - \alpha\frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}}, \tag{6}$$

where α is the learning rate and where the subscript n denotes the optimization step. Furthermore, training stability often depends on appropriate weight initialization strategies, including the Glorot & Bengio (2010) and He *et al.* (2015) initialization methods.

2.2.3 Complex-valued neural networks

RVNNs can process complex-valued data by treating the real and imaginary components as independent inputs. However, this approach may lead to losing phase relationships and requires twice as many parameters to represent complex-valued transformations (Hirose 2012). CVNNs directly operate in the complex domain to address these limitations. In contrast to an RVNN, a CVNN approximates the output of a function $\mathbf{y} \in C$ given a set of input

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variables $\mathbf{x} \in C$. The mathematical formulation is similar to that of an RVNN, except that weights and biases are complex-valued, that is, $\boldsymbol{\theta} = \boldsymbol{\theta}' + i\boldsymbol{\theta}''$.

Various activation functions can be used in CVNNs (Bassey *et al.* 2021; Lee *et al.* 2022). These include fully complex functions, such as tanh, and rectangular split-type functions, which apply separate activation functions to the real and imaginary components, as in

$$\psi(z) = \psi(z') + i\psi(z'') \quad \text{with } z \in C. \tag{7}$$

Activation functions for CVNNs require careful selection, as standard real-valued functions may not preserve complex differentiation. Moreover, Liouville's theorem makes it impossible to have a bounded and holomorphic complex activation function that is not a constant value (Song *et al.* 2015). Thus, alternatives to fully complex functions and split-type activations exist. For instance, the cardioid activation function constrains outputs within the unit circle, making it useful for phase-sensitive applications (Virtue *et al.* 2017), while the ModReLU function enables phase-preserving transformations by applying a thresholding operation in polar coordinates (Arjovsky *et al.* 2016). CVNNs with two or more hidden layers are universal approximators if the activation function is not a polynomial, holomorphic or antiholomorphic function (Voigtlaender 2023). In contrast, CVNNs with only a single hidden layer are universal approximators only if the real or imaginary part of the activation function is not polyharmonic (Voigtlaender 2023).

2.2.4 Optimization of complex-valued parameters

The optimization of CVNN parameters follows a similar approach to RVNNs but requires modifications due to the complex nature of the weights. The CVNN weights are updated according to

$$\boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}_n - \alpha \frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}^*}, \tag{8}$$

where α is the learning rate and $\boldsymbol{\theta}^*$ denotes the complex conjugate of $\boldsymbol{\theta}$. Unlike RVNNs, where we can compute gradients directly, CVNNs rely on Wirtinger calculus to compute gradients with respect to complex-valued parameters. Specifically, the complex gradient is

$$\frac{\partial}{\partial \boldsymbol{\theta}^*} = \frac{1}{2}\left(\frac{\partial}{\partial \boldsymbol{\theta}'} + i\frac{\partial}{\partial \boldsymbol{\theta}''}\right). \tag{9}$$

This formulation allows for the computation of gradients consistent with complex differentiation. As for weight initialization, the Glorot & Bengio (2010) and He *et al.* (2015) methods must be adjusted for CVNNs due to differences in the neural network input and output sizes (Trabelsi *et al.* 2018).

2.2.5 Neural network equivalence

Each complex-valued parameter in a neural network is equivalent to two real-valued parameters, as the complex plane C is isomorphic to R^2 (Barrachina *et al.* 2021b). Consequently, a CVNN has a theoretical advantage over an RVNN of the same architecture due to its higher representational capacity. One approach to balancing network capacity is to double the number of hidden units in each RVNN layer to match the number of real-valued parameters in a CVNN (e.g. Hirose 2012). However, this method unfairly increases the number of trainable parameters in the RVNN, potentially giving it greater function approximation capacity. Mönning & Manandhar (2018) propose an alternative approach of doubling the number of hidden units in every other layer of an RVNN with an odd number of hidden layers. A more recent strategy ensures that the CVNN retains at least half as many trainable parameters as the RVNN (Jojoa *et al.* 2022), though this requires some trial and error.

Table 1. Input size, output size and activation function used in each experiment for CVNNs and RVNNs.

Exp.	Attribute	CVNN	RVNN
I	Input size	32	64
	Output size	1	1
	Activation	sigmoid_C	sigmoid
II	Input size	32	64
	Output size	6	6
	Activation	cardioid	ReLU
III	Input size	2	2
	Output size	3	6
	Activation	$\tanh_C$	tanh

While the previous approaches attempt to balance trainable parameters, they fail to preserve the aspect ratio of the networks, which is a key consideration when designing equivalent architectures. To address this, Barrachina *et al.* (2021b) suggest fixing all bias vectors to zero and determining the RVNN layer dimensions, $N_R^{(l)}$, using a constant aspect ratio r defined by

$$N_R^{(l)} = r N_C^{(l)}, \tag{10}$$

with $1 \leq r < 2$ and where $N_C^{(l)}$ are the desired CVNN layer dimensions. We refer the reader to Barrachina *et al.* (2021b) for determining the optimal r that ensures the RVNN and CVNN maintain equivalent aspect ratios and capacity based on the input and output data types and sizes.

2.2.6 Neural network operations

In RVNNs, multiplying a weight matrix element $w \in R$ with any input element $z \in R$ occurs in a single operation. In contrast, a CVNN performs complex-valued multiplication, that is,

$$zw = (z'w' - z''w'') + i(z'w'' + z''w') \quad \text{with } z \text{ and } w \in C, \tag{11}$$

requiring four real-valued multiplications and two additions per operation. This additional computational cost makes CVNNs inherently more expensive than their real-valued counterparts. Although we can improve the CVNN efficiency by adopting a polar representation for complex weights and inputs (Mönning & Manandhar 2018), CVNNs still require more operations than RVNNs.

3 METHODS

All neural networks used in this study are multilayer perceptrons. To compare CVNN and RVNN performance across tasks, we define three base architectures: A, B and C. These differ in network depth and width: Architecture A is shallow, with a single hidden layer; Architecture B increases depth with two or three layers depending on the task; and Architecture C is the deepest, with up to four hidden layers. Hidden layer sizes are task-specific and scaled to control model capacity. For each CVNN architecture, we construct an RVNN counterpart with the same number of layers, adjusting the layer widths using the method of Barrachina *et al.* (2021b) to match the total number of real-valued trainable parameters (i.e. capacity) and aspect ratio. Table 1 summarizes the neural network input dimensions, output dimensions and activation functions for Experiments I through III. All CVNN activation functions we use enable

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Table 2. Architecture details for CVNNs used in Experiments I–III. Architectures A, B and C vary by depth and size. Equivalent RVNNs use the same number of layers, but their layer sizes are obtained by multiplying the CVNN layer sizes with the aspect ratio (r). Capacity is the total number of real-valued trainable parameters.

Exp.	Arch.	Layers	Layer size	r	Capacity
I	A	1	4	1.02	264
	B	2	8	1.11	656
	C	4	16	1.28	2592
II	A	1	24	1.09	1824
	B	2	32	1.27	4480
	C	3	40	1.33	9440
III	A	1	32	1.14	256
	B	2	32	1.39	2304
	C	2	64	1.40	8704

universal approximation (Voigtlaender 2023). Table 2 summarizes the number of layers, layer sizes, aspect ratios and capacities of Architectures A, B and C across the experiments. The following sections describe the design of each experiment.

3.1 Experiment I: mineral classification

The first experiment focuses on classifying geological mixtures of pyrite and graphite minerals disseminated within water-saturated quartz sand using their respective SIP responses.

3.1.1 Data model

The perfectly polarized interfacial polarization (PPIP) model provides an analytical formulation for the σ_{eff} of geological mixtures containing conductive mineral inclusions of various types and shapes suspended in an electrolyte (Misra *et al.* 2016). The surface-conductance-assisted interfacial polarization (SCAIP) model extends this formulation to spherical inclusions that are non-conductive, such as quartz and clays. As our data model for Experiment I, we use the combined PPIP-SCAIP model (Jin *et al.* 2019) to simulate the σ_{eff} of geological mixtures containing both conductive and non-conductive spherical inclusions in an electrolytic medium. This joint formulation can represent natural materials such as water-saturated sedimentary rocks containing disseminated graphite and pyrite.

The PPIP-SCAIP model depends on 12 physical parameters grouped into three categories. The five parameters describing the conductive inclusions are the characteristic length a_{i} (m), volume fraction ϕ_{i} (per cent), diffusion coefficient D_{i} (m^2 s^{-1}), conductivity σ_{i} (S m^{-1}) and absolute permittivity $\epsilon_{\text{i}} = \epsilon_0\epsilon_{\text{r,i}}$ (F m^{-1}). ϵ_0 is the permittivity of free space and ϵ_{r} denotes a relative permittivity. The four parameters describing the non-conductive inclusions are the characteristic length a_{q} (m), volume fraction ϕ_{q} (per cent), surface conductance λ (S) and absolute permittivity $\epsilon_{\text{q}} = \epsilon_0\epsilon_{\text{r,q}}$ (F m^{-1}). The three parameters describing the host medium are the diffusion coefficient D_{h} (m^2 s^{-1}), conductivity σ_{h} (S m^{-1}) and absolute permittivity $\epsilon_{\text{h}} = \epsilon_0\epsilon_{\text{r,h}}$ (F m^{-1}).

The complex conductivity response σ_{eff} for a mixture of spherically shaped conductive and non-conductive inclusions in an electrolyte solution, as derived from the PPIP-SCAIP model, is

$$\sigma_{\text{eff}}(\omega) = (\sigma_{\text{h}} + i\omega\epsilon_{\text{h}})\frac{1 + 2\left[\phi_{\text{i}} f_{\text{sph}}^{\text{cond}}(\omega) + \phi_{\text{q}} f_{\text{sph}}^{\text{ncond}}(\omega)\right]}{1 - \left[\phi_{\text{i}} f_{\text{sph}}^{\text{cond}}(\omega) + \phi_{\text{q}} f_{\text{sph}}^{\text{ncond}}(\omega)\right]}, \tag{12}$$

where $f_{\text{sph}}^{\text{cond}}$ and $f_{\text{sph}}^{\text{ncond}}$ are the dipolar field coefficients of the conductive and non-conductive spherical inclusions, respectively. We refer the reader to Appendix A for the full expressions of $f_{\text{sph}}^{\text{cond}}$ and $f_{\text{sph}}^{\text{ncond}}$.

3.1.2 Data generation

We sample the 12 physical parameters of the PPIP-SCAIP model to simulate the SIP responses of mixtures containing either pyrite or graphite as conductive inclusions within a water-saturated quartz sand matrix. The four parameters of the non-conductive inclusions and the three parameters of the host medium are held constant across all samples to emphasize the polarization contrast between pyrite and graphite only. For the five parameters describing the conductive inclusions (a_{i}, ϕ_{i}, D_{i}, σ_{i} and ϵ_{i}), values are sampled within intervals consistent with the literature and the physical constraints of the model. We generate a total of 10 000 synthetic mixtures using Latin hypercube sampling in a log-transformed parameter space, following the approach of Bérubé & Baron (2023), to capture parameter variations spanning multiple orders of magnitude. Table 3 presents the parameter intervals used for the pyrite and graphite inclusions. Table 4 provides the sand matrix and electrolyte parameters.

We calculate σ_{eff} for each synthetic mixture using eq. (12) at 32 excitation frequencies logarithmically spaced between 100 Hz and 1 MHz. This results in 5000 samples corresponding to spherical graphite inclusions and 5000 corresponding to pyrite. Fig. 1 presents the distribution of σ_{eff} spectra for the pyrite and graphite mixtures, showing differences and overlap between the two classes.

3.1.3 Neural network dimensions

In Experiment I, the CVNNs take 32-D complex-valued conductivity vectors as input, while the corresponding RVNNs use 64-D real-valued vectors formed by concatenating the real and imaginary components (Table 1). Both architectures produce a real-valued scalar output used for binary classification. For the CVNN, we use the sum of the real and imaginary parts of its output to get a real-valued scalar. The three CVNN architectures, A, B and C, consist of 1, 2 and 4 hidden layers, respectively. Within each architecture, all hidden layers have the same number of neurons: 4 for A, 8 for B and 16 for C (Table 2). These layer configurations yield aspect ratios of 1.02, 1.11 and 1.28, and result in model capacities of 264, 656 and 2 592 real-valued trainable parameters, respectively.

3.1.4 Activation function

The RVNN activation in Experiment I is the sigmoid function defined by

$$\text{sigmoid}(z) = \frac{1}{1 + \exp(-z)} \quad \text{with } z \in R, \tag{13}$$

whereas the CVNN activation function is a modified sigmoid inspired by the non-linearities used in Hirose (2012) and Arjovsky *et al.* (2016). The modified sigmoid reads

$$\text{sigmoid}_C(z) = \text{sigmoid}(|z|)\frac{z}{|z|} \quad \text{with } z \in C. \tag{14}$$

3.1.5 Loss function

Experiment I is a binary classification task in which the negative and positive classes correspond to SIP responses generated by graphite

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Table 3. Parameter distributions used to generate SIP data for pyrite and graphite mixtures in Experiment I.

Parameter	Unit	Pyrite (min)	Pyrite (max)	Graphite (min)	Graphite (max)
a_i	m	1×10^{-4}	1×10^{-3}	1×10^{-4}	1×10^{-3}
ϕ_i	–	0.01	0.20	0.01	0.20
D_i	$m^2\ s^{-1}$	2.5×10^{-4}	7.5×10^{-4}	7.5×10^{-5}	2.5×10^{-4}
σ_i	$S\ m^{-1}$	2×10^{1}	2×10^{3}	2×10^{3}	5×10^{4}
ϵ_i	ϵ_0	11	13	14	16

Table 4. Fixed host electrolyte and quartz matrix parameters used for all SIP simulations in Experiment I.

Parameter	Unit	Value
a_q	m	10^{-3}
ϕ_q	–	0.70
ϵ_q	ϵ_0	4
D_h	$m^2\ s^{-1}$	10^{-9}
σ_h	$S\ m^{-1}$	0.1
ϵ_h	ϵ_0	70
λ	S	10^{-8}

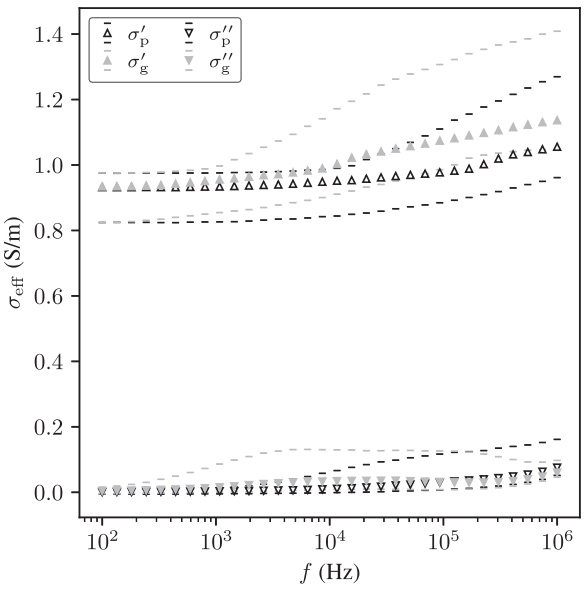

Figure 1. Distribution of σ_{eff} spectra representing pyrite (σ_p) and graphite (σ_g) mixtures for the mineral classification experiment. The triangle markers indicate the distribution mean of 10 000 samples, and the error bars are at the 0.16 and 0.84 quantiles.

and pyrite inclusions, respectively. Therefore, we use the average binary cross-entropy loss, defined as

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{k}^{N}\left[y_k \log p(y_k) + (1 - y_k)\log(1 - p(y_k))\right], \quad (15)$$

where $y_k \in \{0, 1\}$ denotes the target class label for sample k, $p(y_k)$ is the corresponding predicted probability of the positive class and N is the number of samples.

3.2 Experiment II: Cole–Cole inversion

The second experiment evaluates the capacity of RVNN and CVNN models to recover Cole–Cole model parameters from SIP data.

3.2.1 Data model

We use the Pelton complex resistivity formulation of the Cole–Cole equation as our data model for Experiment II. Specifically, we use a sum of two Cole–Cole relaxations reading (Chen *et al.* 2008)

$$\rho_{CC}(\omega) = \rho_0\left[1 - \sum_{k=1}^{2} m_k\left(1 - \frac{1}{1 + (i\omega\tau_k)^{c_k}}\right)\right], \quad (16)$$

where ρ_0 is the resistivity at zero frequency (Ωm), m_k is the dimensionless chargeability, τ_k is the characteristic relaxation time (s) and c_k is the dimensionless Cole–Cole exponent for the kth mode ($k = 1, 2$). This double Cole–Cole formulation allows for modelling multiple overlapping polarization processes commonly observed in natural geomaterials. We normalize the Cole–Cole resistivity spectra by ρ_0 such that the data model depends only on six parameters: m_1, m_2, c_1, c_2, τ_1 and τ_2.

3.2.2 Data generation

We generate 10 000 normalized complex resistivity spectra covering 32 logarithmically spaced frequencies between 10 mHz and 10 kHz by uniformly sampling the six Cole–Cole parameters. The chargeabilities m_1 and m_2 are sampled linearly under the constraint that their sum does not exceed 1. The exponents c_1 and c_2 are drawn uniformly between 0.1 and 1. We deliberately exclude values of $m = 0$ and $c < 0.1$ because small chargeabilities and Cole–Cole exponents yield ill-defined polarization peaks in SIP data and are difficult to recover accurately (Bérubé *et al.* 2017). The relaxation times τ_1 and τ_2 are sampled logarithmically within the excitation frequency band and constrained to differ by at least two orders of magnitude to minimize overlap between the two polarization modes.

3.2.3 Neural network dimensions

In Experiment II, the CVNNs take 32-dimensional complex-valued resistivity vectors as input, while the RVNNs use the equivalent 64-D real-valued vectors formed by concatenating the real and imaginary parts (Table 1). Both networks output six values corresponding to the parameters of the double Cole–Cole model: the RVNN predicts real-valued parameters directly, while the CVNN represents them as the complex modulus of its outputs. The three CVNN architectures, A, B and C, consist of 1, 2 and 3 hidden layers, respectively. Within each architecture, all hidden layers have the same number of neurons: 24 for A, 32 for B and 40 for C (Table 2). These layer configurations yield aspect ratios of 1.09, 1.27 and 1.33, and result in model capacities of 1824, 4480 and 9440 real-valued trainable parameters, respectively.

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3.2.4 Activation function

The CVNN activation in Experiment II is the cardioid function, reading (Virtue *et al.* 2017)

$$\text{cardioid}(z) = \frac{z(1+\cos(\arg(z)))}{2} \quad \text{with } z \in C, \tag{17}$$

and we use ReLU for RVNNs, which is defined by

$$\text{ReLU}(z) = \max(0, z) \quad \text{with } z \in R. \tag{18}$$

The cardioid and ReLU functions are equivalent for real-valued inputs.

3.2.5 Loss function

Experiment II is a regression task and relies on the MSE loss defined by

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{k}^{N} |\hat{y}_k - y_k|^2 \quad \text{with } \{y, \hat{y}\} \in R, \tag{19}$$

where $\hat{y}_k$ are the predicted Cole–Cole parameters for sample k, y_k are the corresponding ground truth parameters and N is the number of samples.

3.3 Experiment III: mechanistic modelling

The third experiment addresses the mechanistic modelling of ion transport and electric potential near a polarizable mineral using the Poisson–Nernst–Planck (PNP) equations (Wong 1979). The system is first solved analytically in spherical geometry to characterize ion migration and diffusion under the influence of a transient electric field. We then train neural networks to approximate the PNP solution and assess the capacity of RVNNs and CVNNs to act as universal approximators in SIP modelling.

3.3.1 Data model

The PNP equations describe the coupled behaviour of ionic migration, diffusion and accumulation and the potential difference arising from charge separation in an electrolyte subjected to an electric field. The system governs the perturbed ionic concentrations δn_j and electric potential δU, and reads (Wong 1979; Bücker *et al.* 2018)

$$i\omega \frac{\delta n_j(\mathbf{r}, \omega)}{D_j} = \nabla^2 \delta n_j(\mathbf{r}, \omega) + \frac{e}{k_{\text{B}} T} v_j n_j^{\infty} \nabla^2 \delta U(\mathbf{r}, \omega), \tag{20}$$

$$\nabla^2 \delta U(\mathbf{r}, \omega) = -\frac{F}{\epsilon_0 \epsilon_{\text{r}}} \textstyle\sum_{j=1}^{2} v_j \delta n_j(\mathbf{r}, \omega), \tag{21}$$

where $\mathbf{r}$ is a position vector, D_j is the diffusion coefficient of ion species j ($\text{m}^2\ \text{s}^{-1}$), v_j is its valence, n_j^{∞} is its equilibrium concentration (mol m^{-3}), e is the elementary charge (C) and ∇^2 is the Laplace operator. Temperature is denoted T (K), k_{B} is Boltzmann's constant (J K^{-1}), F is Faraday's constant (C mol^{-1}) and $\epsilon_0 \epsilon_{\text{r}}$ is the absolute permittivity of the medium (F m^{-1}). For simplicity, we assume the system contains only inactive anions and cations ($j = 1$ refers to anions and $j = 2$ refers to inactive cations), that both ion species share the same diffusion coefficient, and that the electrolyte is symmetric and monovalent. We refer the reader to Bücker *et al.* (2018) for the full analytical expressions of δn_j and δU in the case of a spherical polarizable particle subjected to an external electric field.

3.3.2 Data generation

A circular inclusion of radius a is placed at the centre of a circular input domain of radius $3a$ in 2-D Cartesian coordinates. We uniformly sample 10 000 pairs of spatial coordinates $\mathbf{r}$ within this domain and evaluate the corresponding values of δn_j and δU using the analytical solution of the PNP equations. To ensure consistency, we adopt the same physical parameters as in Bücker *et al.* (2018), using $a = 0.1$ μm and an angular frequency of $\omega = 3 \times 10^6$ rad s^{-1} for the applied electric field.

3.3.3 Neural network dimensions

In Experiment III, CVNNs and RVNNs take 2-D input vectors corresponding to Cartesian coordinates $\mathbf{r}$. For the CVNNs, we encode these inputs as $\mathbf{r} \to \mathbf{r} + i\mathbf{r}$. The CVNN output is a set of three complex-valued fields: perturbed anion concentration, cation concentration and electric potential, corresponding to six real-valued outputs in the RVNNs (Table 1). The three CVNN architectures, A, B and C, consist of 1, 2 and 2 hidden layers, respectively. Within each architecture, all hidden layers have the same number of neurons: 32 for A, 32 for B and 64 for C (Table 2). These layer configurations yield aspect ratios of 1.14, 1.39 and 1.40, and result in model capacities of 256, 2304, and 8704 real-valued trainable parameters, respectively.

3.3.4 Activation function

The CVNN activation in Experiment III is a modified tanh function, reading

$$\tanh_C(z) = \tanh(|z|)\frac{z}{|z|} \quad \text{with } z \in C, \tag{22}$$

and we use the standard tanh function for RVNNs.

3.3.5 Loss function

In Experiment III, we use the complex-valued MSE loss reading (Bassey *et al.* 2021)

$$\mathcal{L}_{\text{MSE}_C} = \frac{1}{2N}\sum_{k}^{N} |\hat{y}_k - y_k|^2 \quad \text{with } \{y, \hat{y}\} \in C, \tag{23}$$

where $\hat{y}_k$ is the predicted PNP solution for sample k, y_k is the corresponding ground truth solution and N is the number of samples.

3.4 Neural network training

Before training, we initialize the RVNN weights using the method of Glorot & Bengio (2010) with gain values adjusted to the appropriate activation functions. For CVNNs, the same method is modified by applying a unitary-preserving scaling factor of $1/\sqrt{2}$ to the initialized weights, following Trabelsi *et al.* (2018). White Gaussian noise with 1 per cent relative standard deviation is added to the data in each experiment to improve realism, regularize neural network training, and lessen the significance of KK constraints. The data sets are randomly split, with 80 per cent used for training and 20 per cent for validation. We then train all networks using the Adam optimizer (Kingma & Ba 2015) to minimize the loss functions described above. We train the models for 100, 10 000 and 1000 epochs for Experiments I, II and III, respectively, using a batch size

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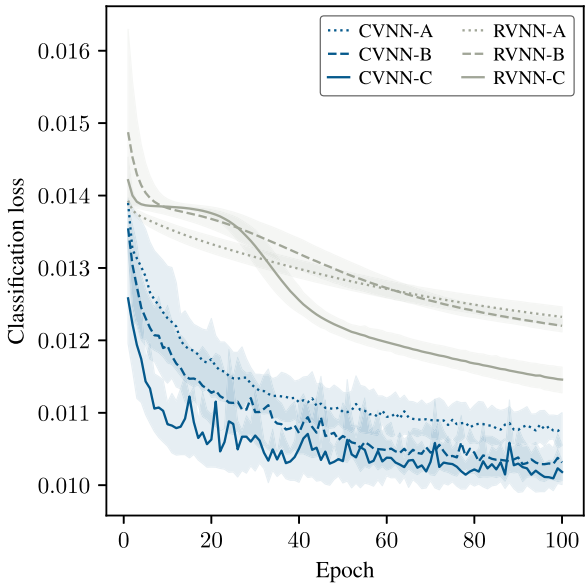

Figure 2. Classification loss of CVNN and RVNN models A, B and C evaluated on the validation data set. Lines are the mean loss across 10 experiment repetitions, and shaded areas are the 0.16–0.84 quantile interval. CVNNs achieve faster convergence and lower classification loss compared to RVNNs across all architectures.

of 50 and an initial learning rate of $\alpha = 10^{-4}$. We repeat each experiment 10 times by regenerating the data set, adding new noise and independently reinitializing and retraining the CVNN and RVNN models.

4 RESULTS

4.1 Experiment I: mineral classification

4.1.1 Learning curves

Fig. 2 shows the classification losses of model configurations A, B and C when evaluated on the validation data set for each of the 100 training epochs. Architecture C achieves the lowest final loss among all models, with CVNN-C converging to 0.0102 and RVNN-C to 0.0115. CVNNs outperform RVNNs for all architectures, with consistent improvements in final loss. For architecture A, CVNN-A reaches 0.0108, compared to 0.0123 for RVNN-A. In architecture B, CVNN-B finishes at 0.0103, whereas RVNN-B settles at 0.0122. Regarding convergence speed, CVNNs minimize loss more rapidly during early training, particularly in architectures B and C. CVNN-C shows a steeper initial descent than RVNN-C and stabilizes earlier around its final value. This result suggests that CVNNs capture salient features of the SIP data more effectively during the initial learning phase, leading to faster training.

4.1.2 Receiver operating characteristic analysis

Fig. 3 presents the receiver operating characteristic (ROC) curves of model architectures A, B and C. Each ROC curve plots the true positive rate against the false positive rate for all possible classification thresholds (in [0, 1]) of the neural network output. Curves approaching the upper-left corner show better classification results, whereas the diagonal line is characteristic of a random classifier.

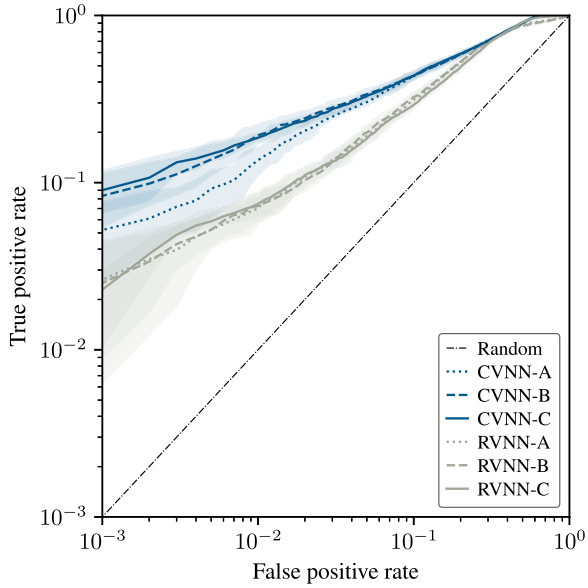

Figure 3. Mineral classification ROC curves of all tested models. Lines are the mean ROC curves across 10 experiment repetitions, and shaded areas are the 0.16–0.84 quantile intervals.

Table 5. ROC AUC for each model architecture in the mineral classification task. Values are the mean ± standard deviation over 10 runs. *p*-values confirm that the advantages of CVNNs over RVNNs are significant.

	A	B	C
CVNN	0.798 ± 0.011	0.801 ± 0.012	0.805 ± 0.008
RVNN	0.755 ± 0.013	0.750 ± 0.004	0.767 ± 0.010
p-value	8.26×10^{-5}	2.86×10^{-7}	1.18×10^{-7}

The CVNN consistently outperforms its RVNN counterpart using all three architectures, with higher true positive rates at nearly every false positive rate threshold. CVNN-C achieves the best performance, followed closely by CVNN-B and CVNN-A. Among the RVNNs, RVNN-C performs best, although still behind all CVNNs. The area under the curve (AUC) values reported in Table 5 confirm this observation. CVNNs outperform RVNNs with all matched architectures, with average ROC AUC margins ranging from 3–5 per cent. CVNN-C yields the highest ROC AUC (0.805 ± 0.008), and RVNN-B obtains the lowest (0.750 ± 0.004). Notably, CVNN-A achieves higher ROC AUC than any RVNN configuration despite its smaller capacity, showing that CVNNs can learn discriminative features in SIP data using less trainable parameters and fewer training epochs.

Overall, the results of Experiment I show that deeper architectures consistently reduce mineral classification loss and improve ROC performance for both CVNNs and RVNNs. CVNNs outperform RVNNs across all model capacities, with the largest ROC AUC gain observed for model B. To assess the statistical significance of the ROC AUC improvements with CVNNs, we perform two-sample *t*-tests on the full set of classification metrics obtained from the validation data set. Each model architecture is evaluated over 2000 validation samples and 10 independent training repetitions, yielding a total of 20 000 ROC AUC metrics per model type. Table 5 reports the resulting *p*-values, which are well below 1 per cent in all cases, indicating that the performance differences between CVNNs and RVNNs are statistically significant.

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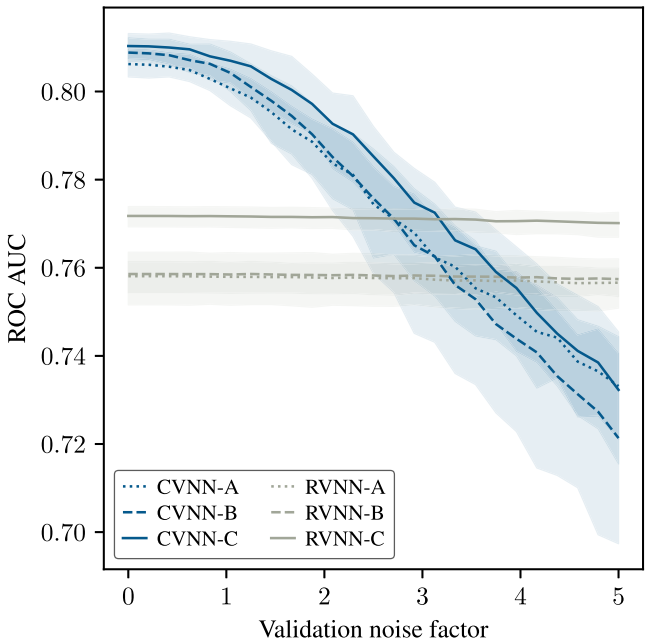

Figure 4. ROC AUC for mineral classification as a function of the validation noise factor. CVNNs outperform RVNNs if the ratio of validation to training data set noise is more than ≈ 3.4.

4.1.3 Noise sensitivity in classification

We analyze the ROC AUC of CVNNs and RVNNs under varying levels of additive Gaussian noise at inference time to evaluate their sensitivity to noise levels exceeding that of the training data. Fig. 4 presents the resulting ROC AUC scores as a function of the noise factor, defined as the ratio between the standard deviations of the validation and training noise levels.

For all architectures, CVNNs achieve higher ROC AUC scores than their real-valued counterparts when the validation noise is equal to or lower than the training noise level (i.e. noise factor ≤ 1). Beyond this threshold, RVNNs gradually become more competitive. For architecture A, CVNN-A outperforms RVNN-A up to a noise factor of approximately 3.4, after which RVNN-A yields slightly better classification metrics. In architectures B and C, the performance crossovers occur closer to a noise factor of 3.2, after which the RVNNs are more robust to unseen noise. These results show that CVNNs leverage the phase structure of SIP data more effectively when noise conditions are stable or moderately perturbed relative to the training set. However, their performance becomes more sensitive to significantly mismatched noise distributions at inference time.

To isolate the impact of mismatched noise levels, we have also evaluated model performance when training and validation data sets contain the same level of noise. In this setup, the classification metric degrades for both model types as noise increases, but CVNNs consistently maintain higher ROC AUC values across all noise levels. This result suggests that CVNNs retain a systematic advantage in classification tasks as long as the noise characteristics during inference are seen during training.

4.2 Experiment II: Cole–Cole inversion

4.2.1 Learning curves

Fig. 5 shows the Cole–Cole parameter estimation loss evaluated on the validation data set after each of the 10 000 training epochs.

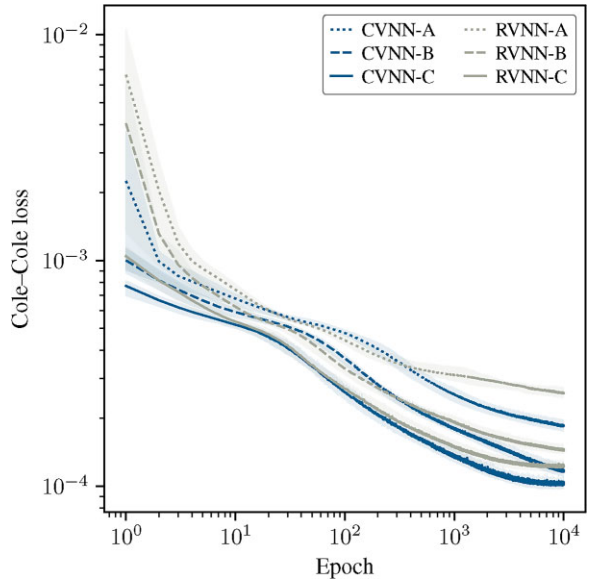

Figure 5. Cole–Cole parameter estimation losses evaluated on the validation data set. Lines and shaded areas are the mean and 0.16–0.84 quantile interval for 10 training repetitions, respectively.

Similarly to the previous experiment, CVNNs outperform RVNNs for all three architectures in terms of final loss. We note that between training epochs 30 and 300, there is a transition phase where the RVNN loss is slightly lower than the CVNN loss. Nevertheless, by the end of the training phase, architecture C achieves the best overall performance, with CVNN-C reaching a final validation loss of 1.0×10^{-4}, compared to 1.2×10^{-4} for RVNN-C. Similarly, architecture B yields final losses of 1.2×10^{-4} (CVNN-B) and 1.5×10^{-4} (RVNN-B), while the shallowest models (architecture A) converge to 1.8×10^{-4} (CVNN-A) and 2.6×10^{-4} (RVNN-A). These results suggest that CVNNs benefit from their ability to preserve the full complex structure of the input SIP data when learning the inverse Cole–Cole model, which may help regularize learning and improve convergence. The final validation loss gap between RVNNs and CVNNs widens in low-capacity models, where complex representations appear to compensate for the limited depth.

4.2.2 Test case on experimental data

As a proof of concept, Fig. 6 compares the predicted Cole–Cole spectra from CVNN-C and RVNN-C to the reference SIP spectrum of metasedimentary rock sample K389055 from the Canadian Malartic disseminated sulphide-associated gold deposit. A comprehensive description of the mineralogical, textural and electrical properties of sample K389055 is available in Bérubé *et al.* (2019). Across the frequency range, the neural networks capture the general dispersion behaviour of the real component of the Cole–Cole response, including the low-frequency plateau and high-frequency decline. As for the imaginary part, the CVNN captures the mid-frequency polarization peak and partially defined high-frequency peak with improved fidelity compared to RVNN. With architecture C, the CVNN shows closer agreement with the true model for ρ' and ρ'', particularly in the low-frequency and high-frequency regions. At intermediate frequencies, RVNN-C underestimates the magnitude of ρ'', while

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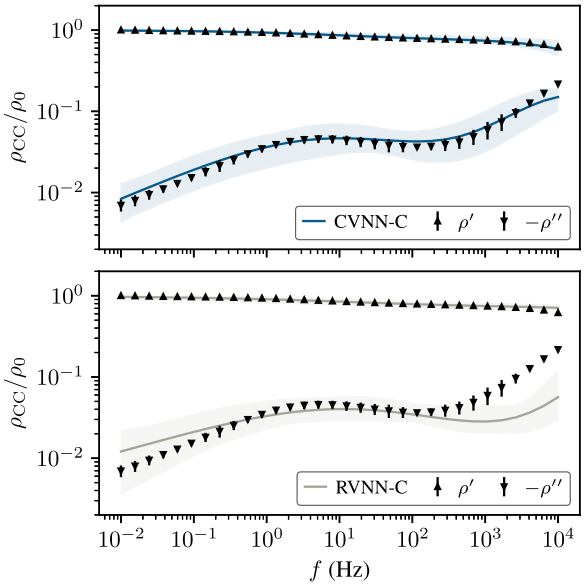

Figure 6. Cole–Cole model fitted to the real part (ρ') and negative imaginary part ($-\rho''$) of the complex resistivity spectra of metasedimentary rock sample K389055, using CVNN-C and RVNN-C models. Predictions are averaged over 10 training repetitions, and shaded regions indicate the 0.16–0.84 quantile interval.

Table 6. NRMSE (in per cent) of Cole–Cole curve fits on the SIP spectra of Canadian Malartic rock samples with the three CVNN and RVNN architectures. Values represent the mean over 10 runs and the uncertainties correspond to one standard deviation. p-values are given for 12 Canadian Malartic rock samples and show that the advantages of CVNNs over RVNNs are significant mostly for ρ''.

	ρ'	ρ''
CVNN-A	24 ± 7	9 ± 2
CVNN-B	21 ± 9	9 ± 1
CVNN-C	20 ± 7	8 ± 2
RVNN-A	15 ± 4	13 ± 2
RVNN-B	18 ± 6	12 ± 2
RVNN-C	20 ± 6	12 ± 1
p-value (A)	7.9×10^{-3}	1.8×10^{-3}
p-value (B)	4.3×10^{-1}	8.8×10^{-3}
p-value (C)	9.8×10^{-1}	3.1×10^{-4}

CVNN-C better reproduces the amplitude and shape of the polarization peak. The 0.16–0.84 quantile intervals also reveal lower variability in the CVNN results, indicating more consistent training outcomes across repetitions. These trends are consistent with the improved validation loss achieved by CVNN-C compared to RVNN-C and suggest that access to complex-valued arithmetic can enhance the accuracy of Cole–Cole inversion, especially in frequency ranges with evident resistivity dispersion.

Table 6 reports the normalized root mean square error (NRMSE) obtained by comparing the Cole–Cole response predicted by each neural network to the previously published SIP spectra of 12 rock samples from the Canadian Malartic deposit (Bérubé *et al.* 2019). We compute the NRMSE by taking the square root of eq. (19) and normalizing it by the range of the ground truth variable. While errors in ρ' vary between 15 and 24 per cent, CVNNs systematically reduce misfit in ρ'', which mainly conveys polarization processes. For all three architectures, CVNNs outperform or match their RVNN counterparts in ρ'' misfit, with improvements ranging from 3 to 5 per cent. This result is consistent with data misfits shown in Fig. 6. The difference in ρ' NRMSE is subtle: RVNNs slightly outperform CVNNs with Architectures A and B, while both models yield identical errors with Architecture C. Despite higher variance in CVNN predictions, t-test p-values below 1 per cent in every case confirm that differences in ρ'' misfit are statistically significant for all architectures.

4.2.3 Cole–Cole parameter error

Since we do not know the ground truth Cole–Cole parameters of actual rock samples, we evaluate the parameter estimation capabilities of RVNN and CVNN models using the synthetic validation data set. Fig. 7 illustrates the NRMSE for each Cole–Cole parameter estimated by CVNN-A and RVNN-A averaged over 10 training runs. The CVNN outperforms the RVNN on all six parameters, with average errors of 14.1 ± 0.2 per cent versus 16.3 ± 0.3 per cent, respectively. The largest discrepancies occur for the exponent parameters c_1 and c_2, where CVNN-A achieves errors of approximately 13 per cent, while RVNN-A exceeds 16 per cent. Chargeability terms m_1 and m_2 also show notable differences, with CVNN-A staying below 16.5 per cent and RVNN-A approaching 17.8 per cent. Relaxation times τ_1 and τ_2 follow the same trend, with the CVNN consistently reducing estimation error by 1–2 percentage points compared to the RVNN. Among all parameters, c_2 exhibits the largest uncertainty in RVNN-A, with a standard deviation of 1.0 per cent. These parameter error patterns are consistent with the trends observed in the SIP curve misfits of Fig. 6. Parameters m and τ, which control the amplitude and frequency location of the relaxation peak, contribute most to the overall curve shape and thus show the smallest improvement in estimation errors with CVNNs. In contrast, the c parameters modulate the dispersion shape more subtly and benefit from the most significant improvement. The strong link between spectral sensitivity and parameter error suggests that misfits on each side of the relaxation peak of the ρ'' spectrum are the main source of error in the c estimates of RVNNs.

Table 7 summarizes the NRMSE values for each Cole–Cole parameter across all architectures. The CVNN-C model achieves the best overall performance, with an average error of 10.6 ± 0.2 per cent across the six parameters, slightly outperforming RVNN-C at 11.5 ± 0.2 per cent. This trend holds consistently across all architectures: CVNNs outperform RVNNs at each depth level, and deeper architectures yield lower parameter errors for both model types. The most significant performance gap appears in architecture A, where CVNN-A outperforms RVNN-A by more than 2 per cent on average, suggesting that the CVNNs may provide better inductive bias in lower-capacity regimes. The p-values for the parameter estimation experiment are well below 1 per cent, indicating statistically significant improvements when using CVNNs. One notable exception is the marginal improvement (p-value $= 2.5$ per cent) for τ_1 when using CVNN-C.

4.2.4 Noise sensitivity in inversion

Fig. 8 presents the total NRMSE for Cole–Cole parameter inversion as a function of the noise level ratio between the validation and training data sets. Across all three architectures, CVNNs outperform RVNNs when validation noise is low, but this advantage diminishes and eventually reverses as the noise level increases.

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Table 7. NRMSE (in per cent) of Cole–Cole parameters estimated with the three CVNN and RVNN architectures. Values represent the mean over 10 runs, and the uncertainties correspond to one standard deviation. p-values are given for 2000 validation samples and show that the advantages of CVNNs over RVNNs are significant.

	m_1	m_2	τ_1	τ_2	c_1	c_2	All
CVNN-A	16.4 ± 0.5	15.7 ± 0.3	13.9 ± 0.2	12.3 ± 0.2	13.0 ± 0.5	13.2 ± 0.5	14.1 ± 0.2
CVNN-B	13.6 ± 0.5	13.3 ± 0.5	11.5 ± 0.3	9.7 ± 0.4	10.0 ± 0.3	10.2 ± 0.3	11.4 ± 0.3
CVNN-C	12.4 ± 0.3	12.2 ± 0.3	11.2 ± 0.2	9.4 ± 0.3	9.4 ± 0.3	9.2 ± 0.3	10.6 ± 0.2
RVNN-A	17.8 ± 0.3	17.5 ± 0.5	15.3 ± 0.4	14.2 ± 0.6	16.1 ± 0.6	16.8 ± 1.0	16.3 ± 0.3
RVNN-B	14.3 ± 0.2	13.8 ± 0.3	12.3 ± 0.3	10.8 ± 0.3	11.6 ± 0.4	12.0 ± 0.3	12.5 ± 0.2
RVNN-C	13.3 ± 0.2	12.9 ± 0.3	11.4 ± 0.3	10.0 ± 0.3	10.4 ± 0.5	11.0 ± 0.4	11.5 ± 0.2
p-value (A)	4.4×10^{-5}	1.1×10^{-5}	3.9×10^{-5}	2.2×10^{-5}	1.2×10^{-7}	3.3×10^{-6}	1.7×10^{-7}
p-value (B)	5.9×10^{-3}	4.5×10^{-2}	8.9×10^{-5}	2.3×10^{-5}	7.4×10^{-6}	9.0×10^{-6}	1.9×10^{-5}
p-value (C)	1.4×10^{-4}	3.7×10^{-4}	2.5×10^{-1}	1.5×10^{-4}	3.6×10^{-4}	6.0×10^{-10}	1.6×10^{-7}

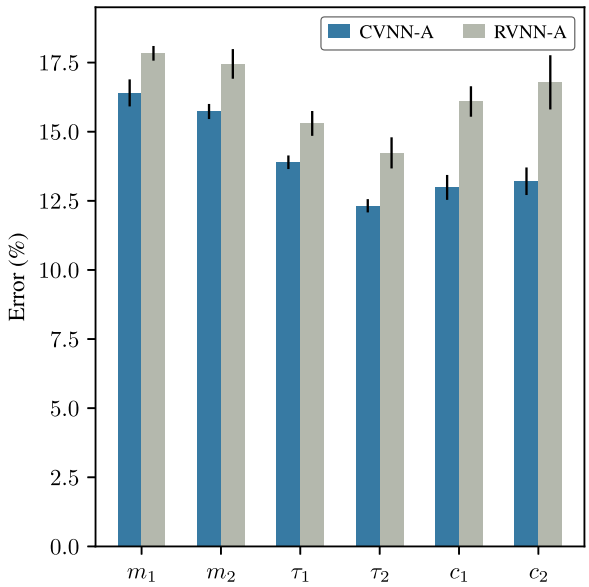

Figure 7. NRMSE of Cole–Cole parameter estimates using 10 training runs of architecture A. Values represent the mean over 10 runs and the error bars represent one standard deviation.

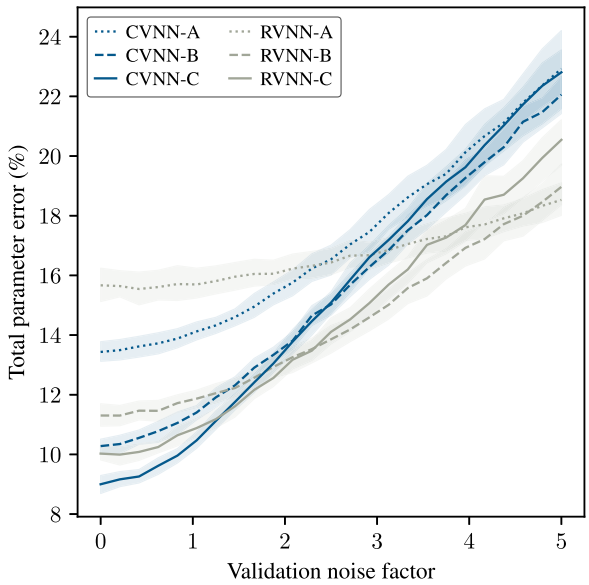

Figure 8. Total NRMSE of Cole–Cole parameter estimates under increasing levels of unseen noise. The validation noise factor is the ratio between the standard deviations of validation and training data sets. Lines are the mean error curves across 10 experiment repetitions, and shaded areas are the 0.16–0.84 quantile intervals.

CVNN-A shows a lower estimation error than RVNN-A when the validation noise factor is below ≈ 2.5, but RVNN-A becomes more accurate once noise exceeds this threshold. For architectures B and C, the crossover occurs for a validation noise factor of ≈ 1.4. Importantly, in all three cases, the crossover occurs above the unit validation noise factor, indicating that CVNNs generalize better within the noise regime seen during training. In contrast, RVNNs are more robust only when exposed to significantly noisier (i.e. up to 2.5 times) out-of-distribution inputs.

To complement this analysis, we conducted a separate experiment in which both the training and validation data were superimposed with matching noise levels ranging from 0 to 5 per cent. In that case, although the total parameter estimation error increased steadily with noise, the performance gap between CVNNs and RVNNs remained stable and consistently in favour of the CVNNs across all architectures. This result indicates that CVNNs are not inherently more sensitive to noise but rather to inconsistencies between training and validation data models.

4.3 Experiment III: mechanistic modelling

4.3.1 Learning curves

Fig. 9 shows the evolution of the modelling loss evaluated on the validation data set for each of the 1000 training epochs. For both model types, deeper architectures consistently yield better performance, with architecture C achieving the lowest final losses and architecture A the highest. CVNNs outperform their RVNN counterparts using all architectures: CVNN-A reaches a final loss of 8.6×10^{-5} compared to 1.32×10^{-4} for RVNN-A, while CVNN-B and CVNN-C achieve 5.9×10^{-5} and 5.8×10^{-5}, respectively—each lower than the corresponding RVNN models, which converge to 7.3×10^{-5} and 7.6×10^{-5}. These results demonstrate the benefit of model depth and confirm that CVNNs offer improved convergence in this function approximation task.

4.3.2 Prediction of perturbation fields

Fig. 10 shows the spatial distributions of the perturbed ionic concentrations and electric potential in response to an external electric field applied from left to right, as predicted by the CVNN-C. The top row shows the real parts of each field, and the bottom row shows the imaginary components. The results reflect the expected physical behaviour described by the analytical model of Bücker *et al.* (2018), including symmetric ionic concentration gradients and a dipolar potential distribution aligned with the applied field. Both the real and imaginary parts are smooth, continuous and free

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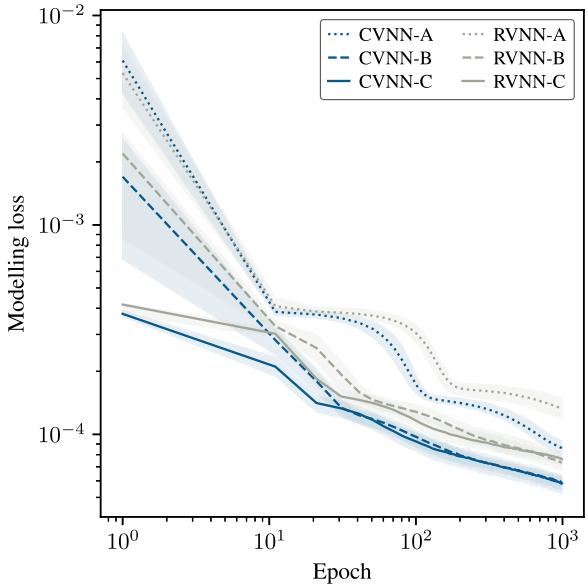

Figure 9. Mechanistic model approximation losses evaluated on the validation data set. Each curve shows the mean validation loss over 10 runs, and shaded regions indicate the corresponding 0.16–0.84 quantile interval.

of spurious artifacts. The boundary conditions are also respected: the potential vanishes at the particle surface, and ionic perturbations vanish at large distances. These observations suggest that the CVNN effectively captures the PNP system's underlying physics. Qualitative inspection of RVNN outputs (not shown) indicates a similar spatial structure, though minor boundary accuracy and smoothness deviations are more apparent when compared with CVNN results.

4.3.3 Modelling error analysis

Fig. 11 provides a comparative overview of the average NRMSE for each physical quantity predicted by CVNN-B and RVNN-B. Overall, the CVNN yields consistently lower errors across all variables. We observe the largest differences in the perturbed ionic concentrations, where the CVNN outperforms the RVNN on both the real and imaginary components. Potential field errors are also reduced for the CVNN, although the performance gap is narrower than ionic concentrations. In particular, both models predict the real component of the potential with high accuracy. The imaginary potential is the most challenging to approximate, likely due to its weaker amplitude in comparison to other variables.

Table 8 reports the NRMSE for each output variable and model combination. In all cases, CVNNs outperform RVNNs, with lower errors for each variable and architecture. Mean errors consistently decrease with depth. Architecture C achieves the lowest average NRMSE for both models, reaching 2.7 per cent for CVNN-C and 3.1 per cent for RVNN-C. The shallowest models show the largest errors, particularly for RVNN-A, whose average error exceeds 4 per cent. The largest discrepancies between CVNN and RVNN performance occur in the ionic concentration components, where CVNN-B yields up to 0.5 per cent lower error than RVNN-B. Overall, $\delta U''$ remains the most difficult to approximate, with errors near 4 per cent for all model types. Standard deviations are generally small, suggesting stable convergence. These results confirm that model depth improves approximation fidelity and that CVNNs consistently outperform RVNNs in this mechanistic modelling task. However, the performance gap becomes more narrow as the models' capacity increases.

The mechanistic modelling results confirm that both CVNNs and RVNNs can accurately approximate the complex-valued PNP equations when trained on the analytical solution. Deeper architectures substantially reduce prediction errors, with CVNNs consistently outperforming RVNNs across all output variables. CVNNs outperform their real-valued counterparts by up to 0.7 per cent lower total NRMSE. For the electric potential, both models achieve comparable accuracy, particularly in deeper configurations. We evaluate the statistical significance of the performance differences between CVNNs and RVNNs by performing two-sample t-tests. Each test aggregates the NRMSE values across 2000 validation samples and 10 independent training runs for each architecture (A, B and C). Table 6 reports the resulting p-values for each predicted perturbation field. In most cases, p-values are well below the 1 per cent threshold and indicate that the NRMSE reductions achieved with all CVNN architectures are statistically significant. However, $\delta U'$ and $\delta U''$ are exceptions for which the improvement using the deeper architectures is marginal with p-values of 1.0–3.6 per cent.

5 DISCUSSION

The results of this study offer a nuanced answer to whether CVNNs are worth adopting for SIP applications. When considering classification, parameter estimation, or forward modelling tasks, CVNNs show modest but statistically significant benefits in comparison with RVNNs when working with laboratory or synthetic SIP data. CVNNs naturally align well with the underlying physics of SIP and enable interpretable inputs and outputs as conduction and polarization processes. Additionally, CVNNs can achieve higher accuracy with smaller model capacity and using fewer training iterations. However, they do so at the cost of increased computational cost, sensitivity to activation functions and unseen noise, and less mature open-source tooling and conventions. For high-value machine learning applications in SIP, where phase preservation and accuracy may be crucial, we recommend using CVNNs despite the extra necessary steps to implement them. RVNNs, on the other hand, provide a simpler and more efficient approach for rapid prototyping or deployment at scale at the cost of lower accuracy.

5.1 Training data volume and number of epochs

While we fixed the number of training samples to 8000 across all experiments to ensure fair comparisons between architectures, we acknowledge that data set size is an important factor in model optimization and generalization. In the case of synthetic data, where we have the possibility to generate an infinite number of labelled examples, training data volume can be scaled arbitrarily. In fact, for sufficiently large data sets, training for a single epoch may be sufficient to reach optimal performance (e.g. Bérubé & Bérubé 2022). Future work could explore the trade-off between data set size and number of epochs for SIP applications by systematically varying the number of training samples and identifying when performance plateaus.

In this study, we chose to fix the number of training epochs across all experiments to provide a consistent comparison between CVNN and RVNN convergence behaviour. Although the validation curves for RVNNs in Experiment I continue to decrease slowly by the end of training (Fig. 2), the CVNN-C loss plateaus after

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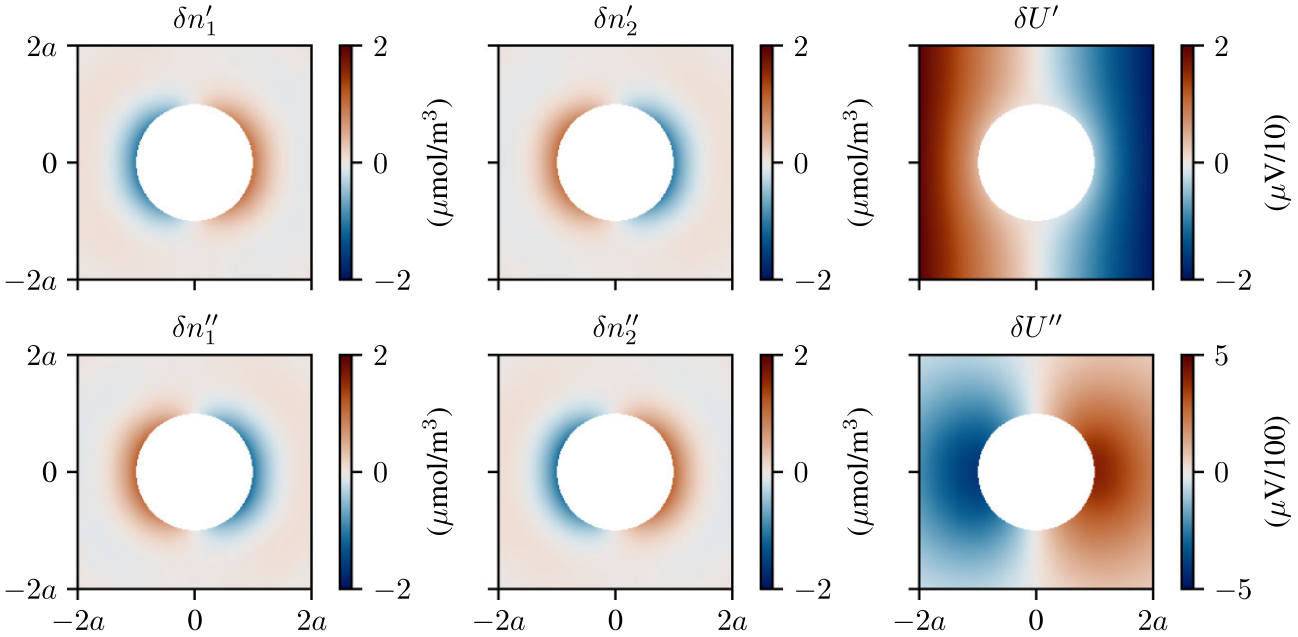

Figure 10. Predicted spatial distributions of CVNN-C for the perturbed ionic concentrations (δn_1 are anions and δn_2 are inactive cations) and perturbed electric potential (δU) around a spherical metallic inclusion of radius $a = 0.1$ µm. An external electric field is applied from left to right. The top row shows the real parts of δn_1, δn_2 and δU; the bottom row shows the corresponding imaginary parts.

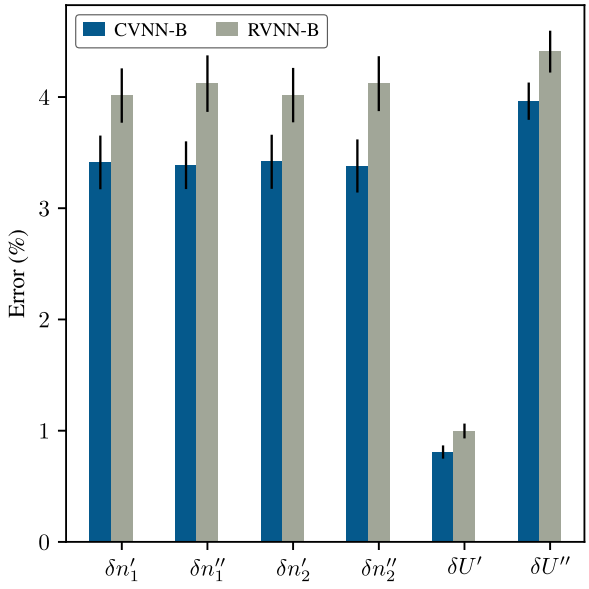

Figure 11. NRMSE of ion concentration and electric potential perturbation fields with predicted with architecture B. Values represent the mean over 10 runs, and the error bars are the 0.16–0.84 quantile interval.

only 40 epochs, whereas the RVNN-C loss has yet to do so after 100 epochs. Although RVNN-C may eventually converge to a comparable loss, its prolonged training time highlights a significant efficiency drawback relative to CVNN-C. In Experiments II and III, this effect is more clearly visualized using logarithmic axes (Figs 5 and 9). In Experiment III, for example, the CVNN-A validation loss decreases from approximately 6×10^{-3} to 2×10^{-4} during the first 10 per cent of epochs, with only marginal improvements from approximately 2×10^{-4} to 1×10^{-4} during the final 90 per cent of epochs. These results support the hypothesis that CVNNs tend to converge faster than RVNNs under matched conditions. While adaptive strategies such as early stopping could further optimize individual training runs, our fixed-epoch design ensures that differences in performance reflect architectural efficiency rather than stopping criteria.

5.2 Complex activation functions

The choice of activation functions strongly influences the results of this study. To ensure fair comparisons between CVNNs and RVNNs, we selected activation functions that are as functionally similar as possible, such as comparing ReLU (RVNN) with the cardioid activation (CVNN). However, these pairings are not unique, and different combinations can significantly alter model performance. The issue is particularly relevant for CVNNs, whose approximation capabilities are more sensitive to the choice of activation due to the mathematical constraints of complex analyticity (Voigtlaender 2023). For example, activation functions that preserve phase structure, such as cardioid and the sigmoid_C and $\tanh_C$ functions we used in the experiments, proved beneficial in all tasks involving laboratory SIP data and synthetic modelling. We observed that rectangular split-type activations were less effective than phase-preserving ones. Moreover, we avoided using the standard tanh and sigmoid functions in CVNNs, even though they naturally map complex inputs to complex outputs, because their singularities led to numerical instabilities and caused training to fail after a few iterations. Future work should investigate a broader range of activations, including learnable and adaptive nonlinearities (e.g. Hirose 2012; Arjovsky *et al.* 2016), that might better capture the phase and amplitude coupling in SIP data.

5.3 Robustness to unseen noise

Our experiments also highlight differences in how CVNNs and RVNNs respond to noise levels not encountered during training. When evaluated on validation data containing significantly higher

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Table 8. NRMSE (in per cent) of the mechanistic model variables estimated with CVNN and RVNN. Values represent the mean over 10 runs, and the uncertainties correspond to one standard deviation. p-values confirm that CVNNs provide a significant advantage in modelling the ionic perturbations.

	$\delta n_1'$	$\delta n_1''$	$\delta n_2'$	$\delta n_2''$	$\delta U'$	$\delta U''$	All
CVNN-A	3.8 ± 0.3	3.8 ± 0.2	3.8 ± 0.2	3.9 ± 0.2	0.92 ± 0.08	4.0 ± 0.2	3.4 ± 0.2
CVNN-B	3.0 ± 0.2	2.9 ± 0.2	3.0 ± 0.2	2.9 ± 0.2	0.70 ± 0.04	3.9 ± 0.2	2.7 ± 0.2
CVNN-C	3.0 ± 0.2	2.9 ± 0.2	3.0 ± 0.2	2.9 ± 0.2	0.66 ± 0.02	3.9 ± 0.2	2.7 ± 0.1
RVNN-A	4.5 ± 0.3	4.8 ± 0.3	4.6 ± 0.3	4.8 ± 0.3	1.24 ± 0.09	4.8 ± 0.2	4.1 ± 0.2
RVNN-B	3.5 ± 0.2	3.4 ± 0.2	3.4 ± 0.2	3.4 ± 0.2	0.76 ± 0.05	4.1 ± 0.2	3.1 ± 0.2
RVNN-C	3.4 ± 0.1	3.4 ± 0.1	3.4 ± 0.1	3.4 ± 0.1	0.77 ± 0.03	4.1 ± 0.1	3.1 ± 0.1
p-value (A)	5.2×10^{-5}	3.9×10^{-6}	3.3×10^{-5}	5.2×10^{-6}	4.6×10^{-5}	1.7×10^{-5}	7.2×10^{-6}
p-value (B)	2.5×10^{-5}	1.9×10^{-5}	3.6×10^{-5}	9.3×10^{-5}	3.4×10^{-2}	3.6×10^{-2}	4.3×10^{-5}
p-value (C)	1.5×10^{-3}	3.8×10^{-5}	3.5×10^{-4}	6.4×10^{-5}	1.1×10^{-5}	1.0×10^{-2}	1.6×10^{-4}

noise than the training set, RVNNs are generally more robust. This result is particularly evident in the parameter inversion task (Experiment II), where performance crossover occurs once validation noise exceeds 1.2–2.0 times the training noise, depending on architecture. In Experiment I, CVNNs outperform RVNNs as long as the validation data noise is less than 3.4 times that of the training data. However, when training and validation noise levels are equal, CVNNs retain their advantage across all tested noise conditions. These findings suggest that while CVNNs more effectively exploit spectral and phase structure in stable noise environments, they may generalize less reliably to previously unseen noise distributions. We recommend considering this sensitivity when deploying CVNNs in SIP applications, where noise levels may vary from those used during model training.

In Experiment I, the near-constant ROC AUC for RVNNs under increasing validation noise factor may reflect the nature of the metric and the model's reliance on amplitude features. ROC AUC evaluates the ranking of predictions rather than absolute values, making it less sensitive to noise that affects the pyrite and graphite classes similarly. Moreover, since RVNNs process real and imaginary components separately, they tend to rely more on the real part of SIP data because the imaginary part is typically much smaller (e.g. Fig. 1). This bias implicitly pushes RVNNs toward learning magnitude-based features, particularly given the small phase angles typical of SIP. In contrast, CVNNs process real and imaginary parts jointly through complex-valued operations, which provides them with a more balanced sensitivity to real and imaginary inputs.

Let a complex-valued SIP signal (eq. 2) be perturbed by additive Gaussian noise such that

$$\tilde{\sigma}_{\text{eff}} = (\sigma' + \delta') + i(\sigma'' + \delta''), \tag{24}$$

where δ' and δ'' are independent zero-mean Gaussian noise terms added to the real and imaginary components, respectively. The perturbed magnitude is then given by

$$|\tilde{\sigma}_{\text{eff}}| = \sqrt{(\sigma' + \delta')^2 + (\sigma'' + \delta'')^2}. \tag{25}$$

By the first-order delta method, the variance of the magnitude is

$$\text{Var}(|\tilde{\sigma}_{\text{eff}}|) \approx \frac{\sigma'^2 \text{Var}(\delta') + \sigma''^2 \text{Var}(\delta'')}{|\sigma|^2}, \tag{26}$$

indicating that magnitude perturbations scale with the inverse of $|\sigma|^2$. The perturbed phase is

$$\arg(\tilde{\sigma}_{\text{eff}}) = \tan^{-1}\left(\frac{\sigma'' + \delta''}{\sigma' + \delta'}\right), \tag{27}$$

and using the first-order delta method, the variance of the phase becomes

$$\text{Var}(\arg(\tilde{\sigma}_{\text{eff}})) \approx \frac{\sigma'^2 \text{Var}(\delta'') + \sigma''^2 \text{Var}(\delta')}{|\sigma|^4}. \tag{28}$$

Unlike the magnitude, phase error scales with the inverse of $|\sigma|^4$, leading to significant amplification of noise effects, especially when $|\sigma|$ is small. Thus, the magnitude features typically exploited by RVNNs are more robust to noise, whereas phase information propagated in CVNNs is more sensitive but potentially more informative when reliable.

5.4 Computational cost

CVNNs inherently require greater computational resources than their real-valued counterparts due to the increased cost of complex-valued operations, including matrix multiplications, element-wise arithmetic and nonlinear activations, all of which require additional real-valued computations (see eq. 11, for example). In our experiments, CVNNs required 140–160 per cent more training time per epoch than RVNNs despite having matched depth, data set size and trainable parameter count. Although this overhead is acceptable for moderate-scale training, it may become prohibitive in large-scale or time-sensitive applications. Notably, CVNNs achieved lower validation losses in fewer epochs, particularly in the mineral classification task (Fig. 2), suggesting that their training efficiency offsets their computational cost. Future improvements in training and inference speed may rely on hardware acceleration for complex arithmetic or algorithmic strategies such as weight sharing (Lecun *et al.* 1998), low-rank approximations (Jaderberg *et al.* 2014) and alternative weight initialization methods in the complex domain (as discussed in Bassey *et al.* 2021).

5.5 Machine learning interpretability

A key strength of CVNNs is their ability to represent physical quantities, such as complex conductivity or perturbed potential, as coherent real and imaginary parts. Our experiments show that CVNNs are advantageous in mechanistic forward modelling, where CVNN outputs mirror the structure of the governing equations. One might ask why approximating forward models with neural networks is a meaningful task. While the mechanistic equations used here describe the response of a single polarizable inclusion of given physical properties, real geomaterials contain millions of such inclusions, each with unique properties. To compute the effective complex conductivity, we must evaluate these models repeatedly and aggregate their responses using effective medium theory. Neural networks trained to approximate the response of a single inclusion can accelerate this process by several orders of magnitude when applied at scale, as shown by Bérubé & Gagnon (2025). Neural networks achieve this acceleration by relying on inherently vectorized matrix multiplications and element-wise activation functions. In contrast,

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analytical solutions to mechanistic SIP models often require evaluating complex nonlinear expressions, and numerical approaches typically involve costly operations such as solving coupled partial differential equations or performing dense numerical integration.

Complex number support is essential to ensure the physical consistency of machine learning models for SIP applications. RVNNs, by contrast, require concatenation or splitting strategies that dilute the phase and amplitude coupling. Considering that SIP data adheres to KK relationships, if the entire frequency spectrum were available and measurements were noiseless, RVNNs could theoretically infer one component from the other as a hidden representation during training, making phase preservation less critical. However, real-world SIP data are typically noisy and bandwidth-limited. All data sets in our experiments include additive noise, and we limit the data sets to a finite frequency range. Notably, Experiment III involved only a single frequency. In such scenarios, CVNNs can explicitly process amplitude and phase information as inputs or outputs with coherent real and imaginary parts. This capability likely accounts for their superior performance across the multiple tasks tested in this study.

6 CONCLUSIONS

This work examines the use of CVNNs by comparing their performance with real-valued models on three representative SIP tasks: mineral classification, Cole–Cole parameter estimation and mechanistic modelling. We perform all comparisons using neural network models with varying architectures and matched capacity to isolate the effect of representational capabilities from model size.

Comparison results show that CVNNs consistently outperform RVNNs in the mineral classification task. CVNNs achieve faster convergence, lower validation loss and higher ROC AUC values than RVNNs, with the largest performance gap observed in shallower networks. However, RVNNs may eventually reach similar predictive performance given more training iterations. In the Cole–Cole inversion task, statistical significance tests confirm that CVNNs outperform RVNNs by achieving lower validation loss, smaller misfit on the imaginary part of resistivity and lower parameter estimation errors for chargeability, relaxation time and Cole–Cole exponent. For the mechanistic modelling task, CVNNs yield significantly lower errors in modelling the ionic concentration fields, particularly in shallow configurations. However, the improvement in imaginary electric potential modelling is marginal.

We contribute with a reproducible benchmark and analysis framework for SIP machine learning and encourage broader adoption and exploration of complex-valued machine learning methods in geophysics. While the performance gains using CVNNs are statistically significant but remain modest, our findings suggest they are most useful when dealing with moderately noisy, phase-sensitive and bandwidth-limited laboratory SIP data. In addition, CVNNs naturally produce the amplitude and phase information necessary to model physical systems describing the SIP effect using PNP equations. While RVNNs may be better suited for low-resource settings, they often require more training iterations and risk missing key data features due to the lack of explicit phase propagation.

This study only tests multilayer perceptrons with simple architectures of one to four layers. SIP applications might benefit from complex-valued convolutional neural networks or recurrent architectures, especially for wideband frequency-domain data. Another promising direction for future research is the development of physics-informed machine learning approaches that explicitly incorporate KK constraints or the PNP equations into the training process. In parallel, further investigation into activation function properties tailored to SIP applications could enhance CVNN performance and stability. Although we test the Cole–Cole inversion scheme on laboratory SIP data, all models in this study are trained on synthetic data. Integrating more experimental data, which typically exhibit higher noise levels and sparser frequency coverage, in the CVNN training phase would help clarify their practical value and limitations beyond the benchmarks considered here. Finally, we emphasize that our experiments are limited to laboratory and synthetic data sets. Further work is required to assess the applicability of CVNNs to field data, where measurement conditions, noise characteristics and bandwidth limitations may pose additional challenges.

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DATA AVAILABILITY

The data and source code used to reproduce the experiments presented in this study are publicly available at https://doi.org/10.5281/zenodo.16950274.

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APPENDIX A: PPIP-SCAIP MODEL COEFFICIENTS

The full derivation of the PPIP-SCAIP model is available in Jin *et al.* (2019). According to the PPIP model, the dipolar field coefficient of the conductive inclusions is

$$f_{\mathrm{sph}}^{\mathrm{cond}}(\omega) = -\frac{1}{2} + \frac{3}{2}\frac{i\omega}{\left[\frac{2}{a_{\mathrm{i}}}\frac{\sigma_{\mathrm{h}}}{\epsilon_{\mathrm{h}}}\frac{E_{\mathrm{h}}}{G_{\mathrm{h}}} - \frac{2}{a_{\mathrm{i}}}\frac{(\sigma_{\mathrm{h}}+i\omega\epsilon_{\mathrm{h}})}{(\sigma_{\mathrm{i}}+i\omega\epsilon_{\mathrm{i}})}\frac{\sigma_{\mathrm{i}}}{\epsilon_{\mathrm{i}}}\frac{F_{\mathrm{i}}}{H_{\mathrm{i}}} + i\omega\left(\frac{2(\sigma_{\mathrm{h}}+i\omega\epsilon_{\mathrm{h}})}{(\sigma_{\mathrm{i}}+i\omega\epsilon_{\mathrm{i}})}+1\right)\right]}, \tag{A1}$$

in which the ratios $F_{\mathrm{i}}/H_{\mathrm{i}}$ and $E_{\mathrm{h}}/G_{\mathrm{h}}$ are

$$\frac{F_{\mathrm{i}}}{H_{\mathrm{i}}} = \frac{a_{\mathrm{i}}[a_{\mathrm{i}}\gamma_{\mathrm{i}} - \tanh(a_{\mathrm{i}}\gamma_{\mathrm{i}})]}{2a_{\mathrm{i}}\gamma_{\mathrm{i}} - (a_{\mathrm{i}}\gamma_{\mathrm{i}})^2\tanh(a_{\mathrm{i}}\gamma_{\mathrm{i}}) - 2\tanh(a_{\mathrm{i}}\gamma_{\mathrm{i}})} \tag{A2}$$

and

$$\frac{E_{\mathrm{h}}}{G_{\mathrm{h}}} = \frac{a_{\mathrm{i}}(a_{\mathrm{i}}\gamma_{\mathrm{h}}+1)}{(a_{\mathrm{i}}\gamma_{\mathrm{h}})^2 + 2a_{\mathrm{i}}\gamma_{\mathrm{h}} + 2}, \tag{A3}$$

where

$$\gamma_{\mathrm{i}}^2 = \left(\frac{i\omega}{D_{\mathrm{i}}} + \frac{\sigma_{\mathrm{i}}}{\epsilon_{\mathrm{i}}D_{\mathrm{i}}}\right) \tag{A4}$$

and

$$\gamma_{\mathrm{h}}^2 = \left(\frac{i\omega}{D_{\mathrm{h}}} + \frac{\sigma_{\mathrm{h}}}{\epsilon_{\mathrm{h}}D_{\mathrm{h}}}\right). \tag{A5}$$

According to the SCAIP model, the dipolar field coefficient of the non-conductive inclusions is

$$f_{\mathrm{sph}}^{\mathrm{ncond}}(\omega) = \frac{Q(R+A) - P}{Q(R-2A) + 2P} \tag{A6}$$

in which

$$A = \frac{1}{a_{\mathrm{q}}^2}, \tag{A7}$$

$$P = \gamma_{\mathrm{h}}^2 + \zeta_{\mathrm{h}}^2\frac{G}{H} + \frac{2G}{a_{\mathrm{q}}^2 L}, \tag{A8}$$

$$Q = \frac{1}{iF+1}\left[2 - \frac{a_{\mathrm{q}}^2\zeta_{\mathrm{h}}^2}{H}\left(\frac{L}{iF} + E\right) - \frac{2E}{L}\right], \tag{A9}$$

$$R = \frac{P}{Q}\left(\frac{iFE+L}{iF+1}\right), \tag{A10}$$

and

$$H = a_{\mathrm{q}}\frac{L_{\mathrm{h}}}{F_{\mathrm{h}}}, \quad G = a_{\mathrm{q}}\frac{G_{\mathrm{h}}}{E_{\mathrm{h}}}, \quad L = \frac{2\lambda}{a_{\mathrm{q}}\sigma_{\mathrm{h}}}, \quad E = \frac{\epsilon_{\mathrm{i}}}{\epsilon_{\mathrm{h}}}, \quad \text{and}$$
$$F = \frac{\omega\epsilon_{\mathrm{h}}}{\sigma_{\mathrm{h}}}. \tag{A11}$$

Finally, the ratios $L_{\mathrm{h}}/F_{\mathrm{h}}$ and $G_{\mathrm{h}}/E_{\mathrm{h}}$ are

$$\frac{L_{\mathrm{h}}}{F_{\mathrm{h}}} = \frac{a_{\mathrm{q}}^2\zeta_{\mathrm{h}}^2 + 2(a_{\mathrm{q}}\zeta_{\mathrm{h}}) + 2}{a_{\mathrm{q}}(a_{\mathrm{q}}\zeta_{\mathrm{h}}+1)} \tag{A12}$$

and

$$\frac{G_{\mathrm{h}}}{E_{\mathrm{h}}} = \frac{a_{\mathrm{q}}^2\gamma_{\mathrm{h}}^2 + 2(a_{\mathrm{q}}\gamma_{\mathrm{h}}) + 2}{a_{\mathrm{q}}(a_{\mathrm{q}}\gamma_{\mathrm{h}}+1)}, \tag{A13}$$

where

$$\gamma_{\mathrm{h}}^2 = \left(\frac{i\omega}{D_{\mathrm{h}}} + \frac{\sigma_{\mathrm{h}}}{\epsilon_{\mathrm{h}}D_{\mathrm{h}}}\right) \tag{A14}$$

and

$$\zeta_{\mathrm{h}}^2 = \frac{i\omega}{D_{\mathrm{h}}}. \tag{A15}$$

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